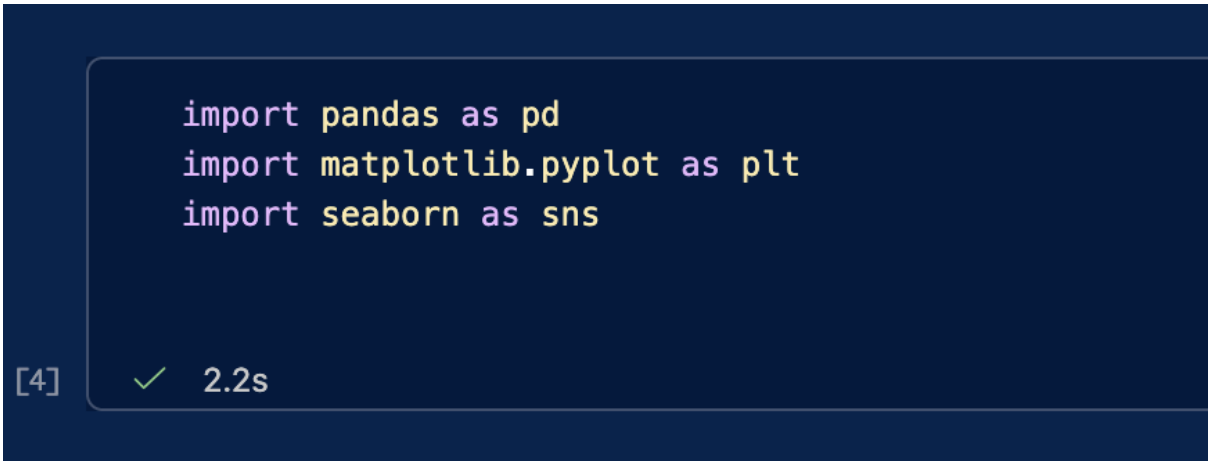
Waist vs AT — Linear Regression

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Cell 1: Importing Libraries

The required libraries, Pandas, Matplotlib, and Seaborn, were imported for data analysis and visualization.



Cell 2: Loading the Dataset

The `WC_AT.csv` dataset was loaded into a Pandas DataFrame using `pd.read_csv()`.



Cell 3: Checking the Dataset

The first five rows, dataset dimensions, and missing values were checked. The dataset contains 109 rows, 2 columns, and no missing values.

```
print(df.head())
print(df.shape)
print(df.isnull().sum())
```

0.0s Python

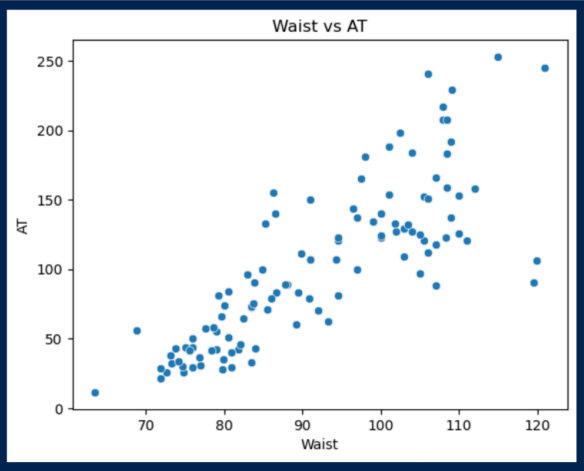
```
   Waist     AT
0  74.75  25.72
1  72.60  25.89
2  81.80  42.60
3  83.95  42.80
4  74.65  29.84
(109, 2)
Waist    0
AT       0
dtype: int64
```

Cell 4: Scatter Plot

A scatter plot was created to visualize the relationship between Waist and AT. The graph shows a positive relationship between the two variables.

```
sns.scatterplot(data=df, x="Waist", y="AT")
plt.title("Waist vs AT")
plt.xlabel("Waist")
plt.ylabel("AT")
plt.show()
```

0.1s Python



Cell 5: Correlation Matrix

The `df.corr()` function was used to calculate the correlation between Waist and AT. The correlation coefficient is **0.818558**, indicating a strong positive relationship.

```
df.corr()
```

[12] ✓ 0.0s

...

	Waist	AT
Waist	1.000000	0.818558
AT	0.818558	1.000000

Cell 6: Linear Regression Model

An Ordinary Least Squares (OLS) regression model was created using Waist as the independent variable and AT as the dependent variable.

```
import statsmodels.formula.api as smf

model = smf.ols('AT ~ Waist', data=df).fit()
```

[9] ✓ 0.1s

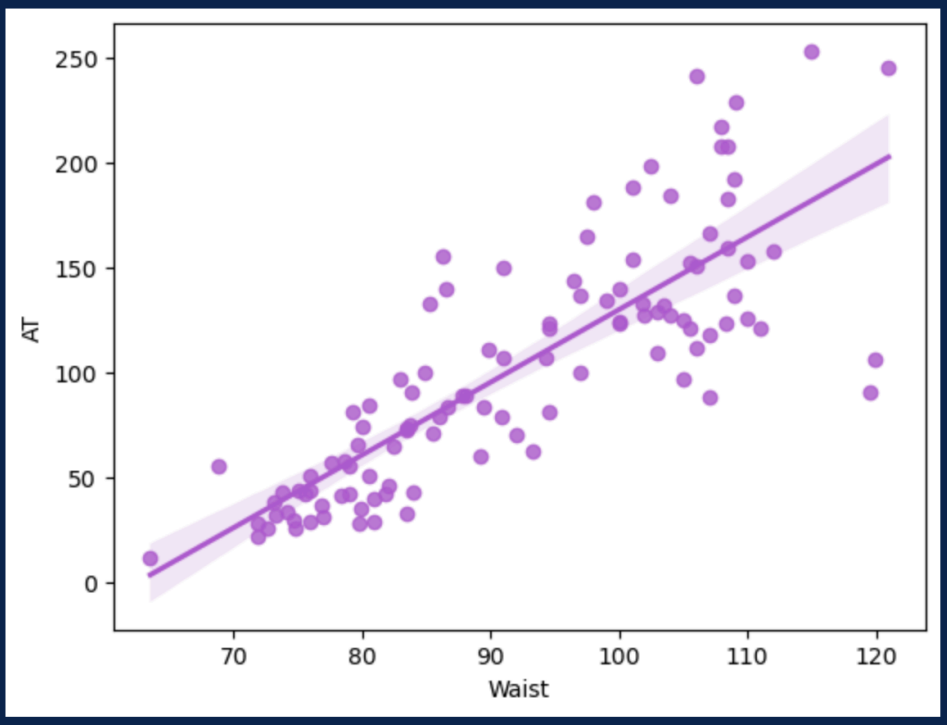
Cell 7: Regression Plot

A regression plot was generated with a fitted line and confidence interval to visualize the relationship between Waist and AT.

```
sns.regplot(
    x="Waist",
    y="AT",
    data=df,
    color="mediumorchid"
)

plt.show()
```

10] ✓ 0.1s



Cell 8: Regression Summary

The OLS summary was displayed to examine the regression coefficients, R-squared value, and statistical significance of the model.

```
model.summary()
```

1] ✓ 0.0s

OLS Regression Results

Dep. Variable:	AT	R-squared:	0.670
Model:	OLS	Adj. R-squared:	0.667
Method:	Least Squares	F-statistic:	217.3
Date:	Tue, 27 Oct 2026	Prob (F-statistic):	1.62e-27
Time:	13:26:12	Log-Likelihood:	-534.99
No. Observations:	109	AIC:	1074.
Df Residuals:	107	BIC:	1079.
Df Model:	1		
Covariance Type:	nonrobust		

	coef	std err	t	P>|t|	[0.025	0.975]
Intercept	-215.9815	21.796	-9.909	0.000	-259.190	-172.773
Waist	3.4589	0.235	14.740	0.000	2.994	3.924

Omnibus:	3.960	Durbin-Watson:	1.560
Prob(Omnibus):	0.138	Jarque-Bera (JB):	4.596
Skew:	0.104	Prob(JB):	0.100
Kurtosis:	3.984	Cond. No.	639.

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

Result

The model achieved an R-squared value of **0.670**, meaning that approximately 67% of the variation in AT is explained by Waist in this dataset. The analysis indicates a statistically significant positive relationship between the two variables.